

Course: B.Tech (T.Y. CSE) Branch : Computer Engineering/Computer Science and Engineering Semester :V

Subject Code & Name: (BTCOC503) Software Engineering

Max Marks: 60

Date: 07/05/25

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		Level/CO	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of these software engineering activities are not a part of software processes	L-2	1
	a. Software development b. Software dependence c. Software validation d. Software specification		
2	The main drawback of the waterfall model is the difficulty of accommodating _____ after the process is underway.	L-1	1
	a. Updation b. Change c. Modification d. None of these		
3	Plan-driven processes are processes where all of the process activities are _____ in advance and progress is measured against this plan.	L-2	1
	a. Planned b. Un Planned c. Planned and un planned d. None of these		
4	How many phases are there in Scrum?	L-2	1
	a. Two b. Three c. Four d. None of these		
5	User requirements are expressed as _____ in Extreme Programming.	L-2	1
	a. Implementation tasks b. Functionalities c. Scenarios d. None of the mentioned		
6	Which one of the following is not a step of requirement engineering?	L-2	1
	a. Elicitation b. Design c. Analysis d. Documentation		
7	Which view in architectural design shows the key abstractions in the system as objects or object classes?	L-2	1
	a. Physical b. Development c. Logical d. Process		
8	_____ & _____ diagrams of UML represent Interaction modeling.	L-1	1
	a. Use Case, Sequence b. Class, Object c. Activity, State Chart d. All of the mentioned		
9	In Design phase, which is the primary area of concern ?	L-2	1
	a. Architecture b. Data c. Interface d. All of the mentioned		

10	Which are open source systems				L-2	1
	a. Apache web server	b. Linux operating system	c. Windows operating System	d. 1 and 2		
11	What are the stages of software testing				L-1	1
	a. User testing	b. Development testing	c. Release testing	d. All of the mentioned		
12	What are the Causes of Software failure				L-1	1
	a. Software failure	b. hardware failure	c. both 1 and 2	d. None of the mentioned		
Q. 2	Solve the following.					12
A)	What are the three general issues that affect many different types of software.				L-1	6
B)	Describe the requirements engineering process using a diagram.				L-2	6
Q.3	Solve the following.					12
A)	Illustrate the Scrum process and describe its key characteristics.				L-2	6
B)	Write a set of functional and non-functional requirements for the MHC-PMS system.				L-3	6
Q. 4	Solve Any Two of the following.					12
A)	Draw a sequence diagram for viewing patient information and explain it in detail.				L-3	6
B)	Discuss three advantages of explicitly designing and documenting software architecture.				L-4	6
C)	When describing a system, explain why you may have to design the system architecture before the requirements specification is complete.				L-3	6
Q.5	Solve Any Two of the following.					12
A)	What are the three fundamental activities of configuration management.				L-1	6
B)	What are the different levels of software reuse.				L-1	6
C)	Draw and explain an object diagram for a Weather Station system				L-3	6
Q. 6	Solve Any Two of the following.					12
A)	What are the steps involved in the fundamental Test-Driven Development (TDD) process? Illustrate with a diagram.				L-2	6
B)	What are the six stages of the acceptance testing process.				L-1	6
C)	Write short note on component and system testing				L-2	6
*** End ***						

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